

Statistics 418: Tools in Data Science

Lecture 1: Class Summary

Class 1: Class Summary

This course is a practical tour of the tools a working data scientist reaches for, and an argument about how to wield them with judgement. Our goal is to build a coherent workflow that runs the full arc of a project: acquiring and understanding data, combining and querying it at scale, modeling it for both *inference* and *prediction*, and communicating what we find. We will work primarily in *R* (through *RStudio*), and we will treat *Claude Desktop* as a genuine member of the toolkit. Along the way, a review class and the midterm consolidate the first half of the course, and a final project, scaffolded by incremental homework, runs as a thread through the entire term.

Class 2: Claude Desktop

AI assistants are rapidly becoming part of the data scientist’s standard toolkit. We will get hands-on with *Claude Desktop* as a workspace for acquiring, transforming, and interrogating data, and (just as importantly) we will practice using it with judgement. A tool that can write code, summarize results, and propose analyses in seconds is enormously powerful, but it will also confidently produce work that is subtly (or grossly) wrong. We will learn to delegate aggressively while verifying relentlessly.

Class 3: Exploratory Data Analysis and Tableau

Exploratory Data Analysis, first memorialized in John Tukey’s classic text of 1977 and now generally known simply as *EDA*, is a paradigm for exploring and understanding data to facilitate efficient, insightful, and accurate analyses. We will explore the context and genesis of EDA and consider how its role and importance has evolved over time. We will also adopt *Tableau* as a tool for rapidly and visually unpacking large, complex data sets.

Class 4: Joins and SQL

Most analyses begin by combining data from multiple sources. We will learn *joins* and *queries* through the lens of *SQL*, the lingua franca of data. We will run SQL directly against our data in *R* using the *squidf* package, which hands our queries to an embedded database behind the scenes. A working fluency in SQL pays dividends everywhere, including in visualization tools like Tableau, which fundamentally “look through the lens” of SQL.

Class 5: Models for Inference

As the old expression goes: “all models are wrong, but some are useful”. Models that allow us to make inferences about processes can be particularly useful, but only if they are accurate and interpretable. Bad news: these conditions can be difficult to satisfy. Worse news: these conditions

can be negatively correlated in that pursuing accuracy can antagonize interpretability and vice versa. We will explore progressively more complex models and stretch the limit of transparency and interpretability as we seek to investigate complex phenomena and draw insights.

Class 6: Models for Prediction

Models that predict can deliver value on a massive scale at limited incremental cost (they do not require an analyst to provide voice-over and interpretation for every case). We will move beyond a toe in the water and build real predictive models in *R*, including *random forests*, while drawing the fundamental contrast between models for *inference* and models for *prediction*. We will also see that predictive models are vulnerable to asymmetric pitfalls, which demand judgement, sophistication, and maturity when developing and deploying them. This sets the stage for a dedicated class on AI and large language models.

Class 7: Review

Having moved quickly through the first half of the course, we will use this class to catch our breath. We will revisit the technical lessons of the first part of the class (joins and SQL, exploratory data analysis, and our first models for inference and prediction), tie the threads together, and make room for the questions that inevitably surface only once the pieces are in place. The goal is to help everyone feel comfortable and confident on the way into the midterm.

Class 8: Midterm

The midterm will include six questions, of which you need complete only five. It is imperative you attend class on the day of the midterm. You will be permitted a single 3x5 card, front and back, as a reference, so prepare it with care. The midterm is worth 20% of your grade.

Class 9: Text Data

Data derived from text can be incredibly valuable, but extracting the data requires a specific set of tools. We will familiarize ourselves with the concept of *regular expressions* and various techniques in *R* for cleaning, parsing, and processing text data, turning unstructured strings into something we can analyze.

Class 10: Artificial Intelligence and Large Language Models

We devote a full class to *large language models* and the broader landscape of modern *AI*. We will build intuition for how these models work (next-token prediction, training and fine-tuning, embeddings) without drowning in the mathematics. Our emphasis will be practical and skeptical: where these tools genuinely create value, where they fail, and why phenomena like *hallucination* make verification a first-class skill rather than an afterthought. This is the natural extension of our contrast between models for inference and models for prediction.

Class 11: From Data to Theory

Having come so far, we will take it back to the top of the class and ask ourselves how we create value in data science all along the spectrum from data to theory. One model for scientific innovation suggests choosing from two paths: 1) present a new and important question, and deliver an intelligent (but perhaps not highly novel) answer, or 2) tackle a classic question and develop a breakthrough technique to solve the problem. A similar framing can be applied in data science: 1) discover or collect profoundly valuable data and present them in a clear and impactful way (perhaps with no analysis at all beyond visualization), or 2) develop new theory and methods that can serve as the building blocks for impactful analyses across a wide range of domains. Here we will explore the spectrum from data to theory and understand how value is created.

Class 12: Final Project

Our final class will be dedicated entirely to your presentations, and it is imperative you attend on the final day. The final project is worth 30% of your grade, which makes it (alongside attendance and participation) the single largest component of your evaluation. Your homework assignments will carry incremental questions, building the project piece by piece, so staying on track throughout the course is the path of least resistance rather than an act of heroic discipline.